ResNet Solved a Problem That Made Deeper Networks Difficult to Train

ARTIFICIAL INTELLIGENCE · MACHINE LEARNING · DEEP LEARNING

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SWIPE ->

Why can't we simply keep adding layers to a neural network and expect it to improve?

One important problem is optimization.

As networks become deeper, training them effectively can become more difficult.

ResNet introduced a powerful idea: the residual connection.

Instead of learning a transformation H(x) directly, a residual block learns a residual function F(x) and adds the original input:

$$H(x) = F(x) + x$$

When dimensions match, the input can bypass the transformation through a shortcut connection.

Why is this useful?

The shortcut provides a direct path for information and gradients to flow through the network.

It also allows a block to learn a residual adjustment rather than having to learn the entire transformation from scratch.

This helped make very deep neural networks practical to train.

ResNet didn't eliminate every optimization challenge, and deeper isn't automatically better.

But residual connections became one of the most influential ideas in deep learning architecture design.

A good architectural innovation doesn't just add complexity.

It makes complexity easier to learn.

#ResNet #DeepLearning #NeuralNetworks

Thanks for reading.

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Code: github.com/yasirusman85/Tank